Derivation of the Light-Shift Hamiltonian Using the Laser Rotating Frame

1 Goal and physical regime

We consider an atom with a set of ground states $\{|g\rangle\}$ and excited states $\{|e\rangle\}$, driven by a monochromatic classical electric field. Our goal is to derive an effective Hamiltonian that acts only within the ground-state manifold when the optical field is far enough detuned that the excited manifold is only virtually populated.

The derivation will follow the sequence

$$\boxed{\text{Schrödinger picture} \longrightarrow \text{laser rotating frame} \longrightarrow \text{RWA} \longrightarrow \text{second-order effective Hamiltonian}} \tag{1}$$

This route is particularly convenient because the rotating-frame Hamiltonian becomes time independent after the rotating-wave approximation (RWA), so ordinary stationary perturbation theory can then be used.

Definitions

D1. $|g\rangle$: a state in the ground-state manifold.
D2. $|e\rangle$: a state in the excited-state manifold.
D3. E_g: energy of $|g\rangle$ in the absence of the optical field.
D4. E_e: energy of $|e\rangle$ in the absence of the optical field.
D5. ω: angular frequency of the applied optical field.
D6. $\hbar$: reduced Planck constant.

2 Bare atomic Hamiltonian

The atomic Hamiltonian is

$$H_A = \sum_g E_g |g\rangle \langle g| + \sum_e E_e |e\rangle \langle e| . \tag{2}$$

We define projectors onto the ground and excited manifolds,

$$P = \sum_g |g\rangle \langle g| , \qquad Q = \sum_e |e\rangle \langle e| , \tag{3}$$

with

$$P + Q = 1, \qquad PQ = QP = 0. \tag{4}$$

3 Classical electric field and dipole interaction

Write the real electric field as

$$\boldsymbol{E}(t) = \frac{1}{2}\left(\boldsymbol{\mathcal{E}}\, \mathrm{e}^{-\mathrm{i}\omega t} + \boldsymbol{\mathcal{E}}^*\, \mathrm{e}^{+\mathrm{i}\omega t}\right), \tag{5}$$

where $\boldsymbol{\mathcal{E}}$ is the complex field amplitude. It can include the polarization vector and its complex phase.

The electric-dipole interaction is

$$V(t) = -\boldsymbol{d} \cdot \boldsymbol{E}(t), \tag{6}$$

where $\boldsymbol{d}$ is the atomic electric-dipole operator.

The full Schrödinger-picture Hamiltonian is therefore

$$H_S(t) = H_A + V(t). \tag{7}$$

4 Raising and lowering parts of the dipole operator

For optical transitions between the ground and excited manifolds, define

$$\boldsymbol{d}^{(+)} \equiv Q\boldsymbol{d}P = \sum_{e,g} \boldsymbol{d}_{eg} \left|e\right\rangle \left\langle g\right|, \tag{8}$$

where

$$\boldsymbol{d}_{eg} \equiv \left\langle e\right| \boldsymbol{d} \left|g\right\rangle. \tag{9}$$

Its Hermitian conjugate is

$$\boldsymbol{d}^{(-)} \equiv P\boldsymbol{d}Q = \left(\boldsymbol{d}^{(+)}\right)^{\dagger}. \tag{10}$$

Thus

$$\boldsymbol{d} \simeq \boldsymbol{d}^{(+)} + \boldsymbol{d}^{(-)}, \tag{11}$$

where direct dipole matrix elements within a single parity manifold are omitted.

Substituting Eqs. (5) and (11) into Eq. (6) gives four terms,

$$\begin{aligned} V(t) = &-\boldsymbol{d}^{(+)} \cdot \boldsymbol{E}^{(+)}(t) - \boldsymbol{d}^{(+)} \cdot \boldsymbol{E}^{(-)}(t) \\ &-\boldsymbol{d}^{(-)} \cdot \boldsymbol{E}^{(+)}(t) - \boldsymbol{d}^{(-)} \cdot \boldsymbol{E}^{(-)}(t), \end{aligned} \tag{12}$$

with

$$\boldsymbol{E}^{(+)}(t) = \frac{1}{2}\boldsymbol{\mathcal{E}}\mathrm{e}^{-\mathrm{i}\omega t}, \qquad \boldsymbol{E}^{(-)}(t) = \frac{1}{2}\boldsymbol{\mathcal{E}}^{*}\mathrm{e}^{+\mathrm{i}\omega t}. \tag{13}$$

5 Transform directly to the laser rotating frame

We now introduce the unitary transformation

$$R(t) = \mathrm{e}^{-\mathrm{i}\omega t Q}. \tag{14}$$

This rotates only the excited manifold at the laser frequency ω.

Define the rotating-frame state by

$$\left|\psi_R(t)\right\rangle = R^{\dagger}(t) \left|\psi_S(t)\right\rangle. \tag{15}$$

The rotating-frame Hamiltonian is

$$H_R(t) = R^{\dagger} H_S(t) R - \mathrm{i}\hbar R^{\dagger}\dot{R}. \tag{16}$$

Because Q commutes with H_A,

$$R^{\dagger} H_A R = H_A. \tag{17}$$

Also,

$$\dot{R} = -\mathrm{i}\omega Q R, \tag{18}$$

so

$$-\mathrm{i}\hbar R^{\dagger}\dot{R} = -\hbar\omega Q. \tag{19}$$

Therefore the bare excited-state energies are shifted according to

$$E_e \longrightarrow E_e - \hbar\omega. \tag{20}$$

6 How the RWA appears in the rotating frame

The rotating transformation acts on the dipole raising and lowering operators as

$$R^\dagger \boldsymbol{d}^{(+)} R = \mathrm{e}^{+\mathrm{i}\omega t} \boldsymbol{d}^{(+)}, \tag{21}$$

and

$$R^\dagger \boldsymbol{d}^{(-)} R = \mathrm{e}^{-\mathrm{i}\omega t} \boldsymbol{d}^{(-)}. \tag{22}$$

Therefore the four products in Eq. (12) transform as

$$\boldsymbol{d}^{(+)} \boldsymbol{E}^{(+)} \longrightarrow \text{constant}, \tag{23}$$
$$\boldsymbol{d}^{(-)} \boldsymbol{E}^{(-)} \longrightarrow \text{constant}, \tag{24}$$
$$\boldsymbol{d}^{(+)} \boldsymbol{E}^{(-)} \longrightarrow \mathrm{e}^{+2\mathrm{i}\omega t}, \tag{25}$$
$$\boldsymbol{d}^{(-)} \boldsymbol{E}^{(+)} \longrightarrow \mathrm{e}^{-2\mathrm{i}\omega t}. \tag{26}$$

Thus the rotating-frame Hamiltonian has the structure

$$H_R(t) = H_{\text{static}} + H_{+2\omega} \mathrm{e}^{+2\mathrm{i}\omega t} + H_{-2\omega} \mathrm{e}^{-2\mathrm{i}\omega t}. \tag{27}$$

The rotating-wave approximation neglects the rapidly oscillating $\mathrm{e}^{\pm 2\mathrm{i}\omega t}$ terms. Physically, these terms average to nearly zero on the slower timescale relevant to the atomic dynamics when

$$|\Omega|,\ |\Delta|,\ \text{ground-state splittings} \ll \omega. \tag{28}$$

7 RWA Hamiltonian in the rotating frame

Define the Rabi-frequency matrix element

$$\Omega_{eg} \equiv \frac{\langle e| \boldsymbol{d} \cdot \boldsymbol{\mathcal{E}} |g\rangle}{\hbar}. \tag{29}$$

With this convention, the factor 1/2 from the real field is not absorbed into Ω_{eg}.

After the RWA, the rotating-frame Hamiltonian becomes

$$\boxed{H_R^{\text{RWA}} = H_A - \hbar\omega Q - \frac{\hbar}{2} \sum_{e,g} \left(\Omega_{eg} |e\rangle \langle g| + \Omega_{eg}^* |g\rangle \langle e| \right)}. \tag{30}$$

This Hamiltonian is time independent.

We now write

$$H_R^{\text{RWA}} = H_{0R} + V_R, \tag{31}$$

where

$$H_{0R} = H_A - \hbar\omega Q, \tag{32}$$

and

$$V_R = -\frac{\hbar}{2} \sum_{e,g} \left(\Omega_{eg} |e\rangle \langle g| + \Omega_{eg}^* |g\rangle \langle e| \right). \tag{33}$$

8 Detuning as a rotating-frame energy difference

Define the optical transition frequency

$$\omega_{eg} \equiv \frac{E_e - E_g}{\hbar}, \tag{34}$$

and the detuning

$$\boxed{\Delta_{eg} \equiv \omega_{eg} - \omega}. \tag{35}$$

The rotating-frame energy of $|e\rangle$ is

$$E_e^{(R)} = E_e - \hbar\omega. \tag{36}$$

Hence

$$\begin{aligned} E_g - E_e^{(R)} &= E_g - (E_e - \hbar\omega) \\ &= -\hbar(\omega_{eg} - \omega) \\ &= -\hbar\Delta_{eg}. \end{aligned} \tag{37}$$

Thus the familiar $1/\Delta_{eg}$ light-shift denominator is simply the inverse energy mismatch between the ground state and the virtually accessed excited state in the laser rotating frame.

9 Why an effective Hamiltonian is needed

Suppose the field is sufficiently far detuned that

$$|\Omega_{eg}| \ll |\Delta_{eg}|. \tag{38}$$

Then the excited manifold is only weakly populated. Nevertheless, the second-order virtual process

$$|g\rangle \longrightarrow |e\rangle \longrightarrow |g'\rangle \tag{39}$$

produces both diagonal energy shifts and off-diagonal couplings within the ground manifold.

Therefore we seek a ground-manifold effective Hamiltonian

$$H_{\text{eff}}^{(g)} = PH_{0R}P + H_{\text{eff}}^{(2)} + O(V_R^3). \tag{40}$$

Because $PQ = 0$,

$$PH_{0R}P = PH_AP. \tag{41}$$

The second-order term is the coherent light-shift Hamiltonian,

$$H_{\text{LS}} \equiv H_{\text{eff}}^{(2)}. \tag{42}$$

10 Systematic second-order block diagonalization

We now derive the general Hermitian second-order effective Hamiltonian rather than guessing its off-diagonal form.

Let

$$H = H_0 + V, \tag{43}$$

where H_0 is diagonal in the retained and eliminated subspaces, and V couples them.

Introduce an anti-Hermitian generator S,

$$S^\dagger = -S, \tag{44}$$

and define

$$\widetilde{H} = \mathrm{e}^{S} H \mathrm{e}^{-S}. \tag{45}$$

Expanding to second order,

$$\widetilde{H} = H_0 + V + [S, H_0] + [S, V] + \frac{1}{2}[S, [S, H_0]] + O(V^3). \tag{46}$$

Choose S such that the first-order coupling between retained and eliminated subspaces vanishes,

$$V + [S, H_0] = 0 \tag{47}$$

for the off-diagonal blocks.

For $|i\rangle$ in the retained subspace and $|m\rangle$ in the eliminated subspace,

$$\langle m| [S, H_0] |i\rangle = (E_i - E_m) S_{mi}. \tag{48}$$

Equation (47) gives

$$S_{mi} = \frac{V_{mi}}{E_m - E_i}, \qquad S_{im} = \frac{V_{im}}{E_i - E_m}. \tag{49}$$

Using $[S, H_0] = -V$, Eq. (46) becomes

$$\widetilde{H} = H_0 + \frac{1}{2}[S, V] + O(V^3). \tag{50}$$

Therefore

$$H_{\text{eff}}^{(2)} = \frac{1}{2} P[S, V]P. \tag{51}$$

For two retained states $|i\rangle$ and $|j\rangle$,

$$\begin{aligned} \langle i| H_{\text{eff}}^{(2)} |j\rangle &= \frac{1}{2} \sum_m (S_{im} V_{mj} - V_{im} S_{mj}) \\ &= \frac{1}{2} \sum_m V_{im} V_{mj} \left[\frac{1}{E_i - E_m} + \frac{1}{E_j - E_m} \right]. \end{aligned} \tag{52}$$

Thus

$$\boxed{\langle i| H_{\text{eff}}^{(2)} |j\rangle = \frac{1}{2} \sum_m V_{im} V_{mj} \left[\frac{1}{E_i - E_m} + \frac{1}{E_j - E_m} \right]}. \tag{53}$$

The symmetrization of the two denominators is essential for Hermiticity when $E_i \neq E_j$.

11 Apply the general formula to the light shift

Now identify

$$|i\rangle = |g'\rangle, \qquad |j\rangle = |g\rangle, \qquad |m\rangle = |e\rangle, \tag{54}$$

and use the rotating-frame unperturbed excited-state energy

$$E_m = E_e - \hbar\omega. \tag{55}$$

From Eq. (33),

$$\langle e| V_R |g\rangle = -\frac{\hbar}{2} \Omega_{eg}, \tag{56}$$

and

$$\langle g'| V_R |e\rangle = -\frac{\hbar}{2} \Omega^*_{eg'}. \tag{57}$$

Thus

$$\langle g'| V_R |e\rangle \langle e| V_R |g\rangle = \frac{\hbar^2}{4}\Omega^*_{eg'}\Omega_{eg}. \tag{58}$$

The denominators are

$$E_g - (E_e - \hbar\omega) = -\hbar\Delta_{eg}, \tag{59}$$

and

$$E_{g'} - (E_e - \hbar\omega) = -\hbar\Delta_{eg'}. \tag{60}$$

Substituting into Eq. (53), we obtain

$$\boxed{\langle g'| H_{\mathrm{LS}} |g\rangle = -\frac{\hbar}{8}\sum_e \Omega^*_{eg'}\Omega_{eg}\left(\frac{1}{\Delta_{eg'}} + \frac{1}{\Delta_{eg}}\right)}. \tag{61}$$

This is the general second-order coherent light-shift Hamiltonian in the ground manifold after the RWA.

12 Diagonal AC Stark shift

For $g' = g$, Eq. (61) becomes

$$\begin{aligned}\langle g| H_{\mathrm{LS}} |g\rangle &= -\frac{\hbar}{8}\sum_e |\Omega_{eg}|^2\left(\frac{1}{\Delta_{eg}} + \frac{1}{\Delta_{eg}}\right) \\ &= -\frac{\hbar}{4}\sum_e \frac{|\Omega_{eg}|^2}{\Delta_{eg}}.\end{aligned} \tag{62}$$

Therefore

$$\boxed{\delta E_g = -\frac{\hbar}{4}\sum_e \frac{|\Omega_{eg}|^2}{\Delta_{eg}}}. \tag{63}$$

With the convention $\Delta_{eg} = \omega_{eg} - \omega$, red detuning corresponds to $\Delta_{eg} > 0$, so the ground-state shift is negative.

13 Hermiticity of the off-diagonal light shift

Taking the complex conjugate of Eq. (61),

$$\begin{aligned}\left[\langle g'| H_{\mathrm{LS}} |g\rangle\right]^* &= -\frac{\hbar}{8}\sum_e \Omega_{eg'}\Omega^*_{eg}\left(\frac{1}{\Delta_{eg'}} + \frac{1}{\Delta_{eg}}\right) \\ &= \langle g| H_{\mathrm{LS}} |g'\rangle,\end{aligned} \tag{64}$$

provided the detunings are real, as they are for the coherent Hamiltonian considered here. Hence

$$\boxed{H^\dagger_{\mathrm{LS}} = H_{\mathrm{LS}}}. \tag{65}$$

A frequently written unsymmetrized form,

$$\langle g'| \widetilde{H}_{\mathrm{LS}} |g\rangle \propto \sum_e \frac{\Omega^*_{eg'}\Omega_{eg}}{\Delta_{eg}}, \tag{66}$$

is not generally Hermitian when $\Delta_{eg} \neq \Delta_{eg'}$. It is only justified when the ground-state splittings are negligible compared with the optical detuning.

14 Nearly degenerate ground manifold

If the ground-state splittings are much smaller than the optical detuning,

$$|E_g - E_{g'}| \ll \hbar|\Delta|, \tag{67}$$

then

$$\Delta_{eg} \simeq \Delta_{eg'} \simeq \Delta_e. \tag{68}$$

Equation (61) reduces to

$$\boxed{\langle g'| H_{\mathrm{LS}} |g\rangle \simeq -\frac{\hbar}{4}\sum_e \frac{\Omega^*_{eg'}\Omega_{eg}}{\Delta_e}}. \tag{69}$$

This is the form commonly used before decomposing the light shift into scalar, vector, and tensor contributions.

15 Dipole-operator form

Using Eq. (29), Eq. (61) can be written as

$$\begin{aligned}\langle g'| H_{\mathrm{LS}} |g\rangle = -\frac{1}{8\hbar}\sum_e \langle g'| \boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}^* |e\rangle \langle e| \boldsymbol{d}\cdot\boldsymbol{\mathcal{E}} |g\rangle \\ \times\left(\frac{1}{\Delta_{eg'}} + \frac{1}{\Delta_{eg}}\right).\end{aligned} \tag{70}$$

In the nearly degenerate ground-manifold approximation,

$$\langle g'| H_{\mathrm{LS}} |g\rangle \simeq -\frac{1}{4\hbar}\sum_e \frac{\langle g'| \boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}^* |e\rangle \langle e| \boldsymbol{d}\cdot\boldsymbol{\mathcal{E}} |g\rangle}{\Delta_e}. \tag{71}$$

16 What was neglected by the RWA?

The discarded rotating-frame terms oscillate as $\mathrm{e}^{\pm 2\mathrm{i}\omega t}$. If they are retained, second-order perturbation theory generates additional counter-rotating contributions with denominators of the form

$$\omega_{eg} + \omega, \tag{72}$$

instead of

$$\omega_{eg} - \omega = \Delta_{eg}. \tag{73}$$

Near resonance,

$$|\omega_{eg} - \omega| \ll \omega_{eg} + \omega, \tag{74}$$

so the counter-rotating contribution is much smaller. The leading correction beyond the RWA is closely related to the Bloch–Siegert shift.

17 Conditions for the derivation

The main assumptions are:

A1. Electric-dipole approximation:

$$V(t) = -\boldsymbol{d}\cdot\boldsymbol{E}(t). \tag{75}$$

A2. Rotating-wave approximation:

$$|\Omega|,\ |\Delta|,\ \text{relevant internal splittings} \ll \omega. \tag{76}$$

A3. Dispersive perturbative regime:

$$|\Omega_{eg}| \ll |\Delta_{eg}|. \tag{77}$$

A4. Coherent Hamiltonian treatment: spontaneous emission is neglected in H_{LS}. If radiative decay is important, the effective dynamics should be written as a master equation containing both a Hermitian light-shift Hamiltonian and dissipative scattering terms.

18 Summary of the logic

The complete derivation can be summarized as follows.

Step 1. Start from

$$H_S(t) = H_A - \boldsymbol{d} \cdot \boldsymbol{E}(t). \tag{78}$$

Step 2. Transform directly to the laser rotating frame using

$$R(t) = \mathrm{e}^{-\mathrm{i}\omega t Q}. \tag{79}$$

Step 3. In that frame, the resonant dipole terms become static, whereas the counter-rotating terms acquire phases $\mathrm{e}^{\pm 2\mathrm{i}\omega t}$.

Step 4. Apply the RWA by neglecting those rapidly oscillating terms.

Step 5. The resulting time-independent Hamiltonian is

$$H_R^{\mathrm{RWA}} = H_A - \hbar\omega Q + V_R. \tag{80}$$

Step 6. The rotating-frame energy mismatch is

$$E_g - (E_e - \hbar\omega) = -\hbar\Delta_{eg}. \tag{81}$$

Step 7. Use the general second-order effective-Hamiltonian formula

$$\langle i|\, H_{\mathrm{eff}}^{(2)}\, |j\rangle = \frac{1}{2}\sum_m V_{im}V_{mj}\left[\frac{1}{E_i - E_m} + \frac{1}{E_j - E_m}\right]. \tag{82}$$

Step 8. Obtain the Hermitian light-shift Hamiltonian

$$\boxed{\langle g'|\, H_{\mathrm{LS}}\, |g\rangle = -\frac{\hbar}{8}\sum_e \Omega^*_{eg'}\Omega_{eg}\left(\frac{1}{\Delta_{eg'}} + \frac{1}{\Delta_{eg}}\right)}. \tag{83}$$

The key conceptual advantage of the rotating-frame approach is that the detuning appears immediately as a static energy difference, while the RWA is visible as the removal of terms oscillating at approximately 2ω. This makes the subsequent second-order perturbation theory especially direct.